Digital Vector · VCE Applied Computing | Unit 1 · Area of Study 1 · Data Analysis · KK09

Design tools for solutions

Imagine being told to build a house. You wouldn't start pouring concrete and hope it works out. You'd get a blueprint first — a plan everyone can read, check and agree on before a single brick is laid.

What this key knowledge covers
This summary distils everything you need for **KK09 — Design tools for solutions** in Data Analysis. Work through the explanations, then use the worked good-vs-weak answers to see exactly how marks are won and lost. Tick off the revision checklist at the end before a SAC or exam.

Study summary

Why would you draw a solution before you build it?

Imagine being told to build a house. You wouldn't start pouring concrete and hope it works out. You'd get a **blueprint** first — a plan everyone can read, check and agree on *before* a single brick is laid.

Data solutions are no different. Before you create a database, a spreadsheet or a data visualisation , you produce **design tools**: small, readable plans that show what the solution will do and what it will look like. They let you (and your teacher, and your client) catch problems while they're still cheap to fix — on paper — rather than after you've spent three lessons building the wrong thing.

A good design tool passes the **blueprint test**: could someone *else* build your solution from it without asking you a single question? If yes, it's detailed enough. If they'd have to guess, it isn't.

Where these tools sit in the problem-solving methodology

Every solution in this course moves through four stages. Design tools live squarely in the second one.

Analysis works out *what* the solution must do (the requirements). **Design** works out *how* it will do it and *what it will look like* — that's where your four tools live. Only then does **Development** build it, and **Evaluation** checks it against the original requirements.

Design tools split neatly into two jobs. Some describe **functionality** — the behaviour, the data flow, the logic (IPO charts and query designs). Others describe **appearance** — the layout, colours and labels a user will see (mock-ups and annotated diagrams). Strong students always know which job each tool is doing.

The four tools at a glance

Each tool answers a different question about your solution. Learn the one-line job of each — it's the difference between naming a tool and explaining it.

A three-column table — **Input · Process · Output** — that maps what data goes in, what is done to it, and what comes out. Plans **functionality**.

A written plan for a question you'll ask the data — which fields, which conditions, which sort/total — for a **database**, **spreadsheet** or **visualisation**. Plans **functionality**.

A sketch of the **layout** — where the title, charts, tables and buttons go on screen or page. A picture of the finished look. Plans **appearance**.

A mock-up (or other diagram) with **callout notes** explaining each design choice — colours, fonts, alignment, chart type — usually tied to **design principles**. Plans & *justifies* appearance.

Try it: sort the design jobs

Drag (or tap, then tap a bin) each job onto the tool that does it. Four belong to each tool — get them home.

IPO charts — input, process, output

An IPO chart is the simplest way to plan what a solution *does*. Three columns, read left to right: what goes **in**, what happens in the **middle**, what comes **out**.

Input = the fruit and milk you tip in. **Process** = the blending. **Output** = the smoothie. You don't see *how* the blades spin — you just describe the transformation. That's exactly the level of detail an IPO chart works at.

Worked example — the StepUp leaderboard

The coordinator wants a leaderboard that ranks home groups by total steps. Here's the IPO chart for that process:

See it move — the IPO machine

Press **Run** to push the raw StepUp data through the machine. Watch the messy input get cleaned, summed and ranked into the output. This is the mental model: data is *transformed*, not just moved.

Mock-ups — a picture of the finished look

A mock-up is a sketch of the *layout*: where the title, charts, tables, buttons and labels will sit. It shows a client what they'll actually see — before you build it.

A mock-up shows the **arrangement** — the furniture in the right places — but the content can be placeholder. Real data isn't needed yet; XX,XXX steps or grey boxes stand in for charts. You're designing the *frame*, not filling it.

Worked example — the StepUp foyer dashboard

Here's a mock-up of the data visualisation that will run on the foyer screen. Notice it's a **layout plan** — blocky, labelled, placeholder content:

Annotated diagrams — a mock-up that explains itself

An annotated diagram is a mock-up (or other diagram) with **callout notes** pointing at each part, explaining the design choice — usually by naming a *design principle* or a *format/convention*.

If a mock-up says "*here is the layout*", an annotated diagram adds "*…and here's **why** each part looks like this*". The annotations are where you show the examiner you understand **design principles** — not just where boxes go.

Query designs — planning the questions you'll ask the data

A query is a question you put to your data: "*which home groups beat 50,000 steps?*". A query **design** plans that question before you build it — naming the fields, the conditions, the sort and any totals.

Queries appear across all three Data Analysis tools, and the design looks a little different in each:

Plan the **fields** to show, the **criteria** (conditions), and the **sort order**. Often written as a QBE grid or an SQL-style statement.

Plan the **filter** conditions and the **functions** that summarise — SUMIF, AVERAGEIF, COUNTIF — and which range they cover.

Plan **which queried fields feed which chart** — e.g. "ranked totals → bar chart", "daily totals → line chart". The query shapes the picture.

Worked example — three StepUp queries

The same data, three different questions, planned three different ways:

Try it: design a database query

Pick the fields, a condition and a sort. Watch the SQL-style design assemble — and the StepLog rows that would match light up. This is the *plan*; the real query comes later in Development.

Appearance vs functionality — what your annotations target

When you annotate a design, you're commenting on one of two things: how it **looks** (appearance) or how it **works** (functionality). Examiners love asking you to do both.

The visual design the user sees. Judged with design principles such as:

The behaviour and usefulness. Judged with factors such as:

Choosing and combining the right tool

No single tool plans a whole solution. A real design uses several together — each covering a part the others can't.

The full StepUp design, tool by tool

A complete design for the StepUp solution combines all four:

Exam traps students fall into

These are the avoidable mistakes that quietly cost marks on design-tool questions. Learn to spot each one in your own answers.

Command words & the P·E·L method

Half of exam success is reading the command word correctly, then structuring your answer so a marker can find every point.

Decode the command word

Name it. No explanation. "It is an IPO chart."

Say what it is *and* a feature/how it works.

Give reasons — *why* or *how*, with a because.

State a clear difference between two things — use "whereas".

Actually *produce the artefact* — the chart, mock-up or query.

Add labelled notes naming principles & reasons.

Argue *why your choice is the best* one.

Weigh strengths and weaknesses, then conclude.

Structure with P·E·L

For "explain / justify" answers, three moves keep you on track:

State your claim directly. "A mock-up plans the layout."

Give the reason or evidence. "…because it shows where each chart sits before building."

Tie back to the scenario / question. "…so the StepUp coordinator approves the look early."

Practice questions — mark yourself

Five VCE-style questions on the StepUp scenario, 18 marks total. Attempt each, reveal the weak vs strong responses and the marking guide, then award yourself a score. Your total tracks in the dock.

- 1 mark — naming each correct tool (max 1 for naming).
- 1 mark — stating what each plans (functionality vs appearance, or a correct specific job).
- Accept IPO, mock-up, annotated diagram, query design.
- 1 mark — Input names the actual fields of the file.
- 1 mark — Process lists clear ordered steps (clean → sum → sort → rank).
- 1 mark — Output is the specific ranked leaderboard (not just "data").
- 1 mark — correct description of a mock-up (layout/appearance).
- 1 mark — correct description of an annotated diagram (mock-up + explanatory notes).
- 1 mark — a clear point of distinction ("whereas…").
- 1 mark — a relevant StepUp example for each.
- 1 mark — correct fields shown (Homeroom + total of Steps).
- 1 mark — the > 50,000 condition (HAVING/criteria).
- 1 mark — sort highest→lowest (ORDER BY DESC).
- 1 mark — a valid reason a query design helps before building.
- 1 mark each (×3) — a clear annotation that names a design principle/factor *and* the reason it helps the user.
- 1 mark — at least one appearance and one functionality annotation present.

- 1 mark — annotations clearly tied to specific parts of the StepUp dashboard.

Key terms

Term	Meaning
Database of step readings	**IPO** data in/clean/store · **Query** top groups
Spreadsheet leaderboard	**IPO** sum & rank · **Query** AVERAGEIF
Dashboard visualisation	**Mock-up** layout · **Annotated** design reasons

Common traps & misconceptions

EXAM Mock-ups are hand-drawn in the exam
You can't open design software in a written exam. If asked to "draw a mock-up", produce a **neat, labelled hand sketch** with a ruler. Markers reward clear regions and labels — not artistic shading. Always note the **orientation** (portrait/landscape) if it matters.

EXAM A query design is not a finished result
Marks are for the **plan** — fields, criteria, sort — written clearly enough to build from. You don't need real output rows. Equally, "find the best home group" is **too vague**: name the field (SUM(Steps)), the condition and the order.

1 Naming a tool instead of using it
"**I would use a mock-up**" is not a mock-up. If the command word is *construct, design, draw or produce*, you must actually **create the artefact**, not describe that you'd make one.

2 Confusing the IPO process column with output
The **process** column describes the *actions* (clean, sum, sort). The **output** is the *result* (the ranked table). Students often put the answer in the process column and leave output blank.

3 Annotations that don't name a principle
"This looks nice" earns nothing. Every annotation must name a **design principle** (contrast, hierarchy, alignment...) or a **convention**, and say *why it helps the user*.

4 Vague query designs
"Find the top group" hides the design. State the **fields**, the **condition** and the **sort** — e.g. SUM(Steps) GROUP BY Homeroom ORDER BY DESC.

5 Mixing up appearance and functionality
If a question asks for an **appearance** annotation, don't write about how the leaderboard updates (that's functionality). Read which one is being asked — sometimes both, clearly separated.

6 Real data in a mock-up
A mock-up plans the **layout**, so use **placeholders** (XX,XXX, grey bars). Filling it with calculated totals turns a design tool into a half-built product and misses the point.

7 Forgetting the chart-type justification
Choosing a **bar** chart to compare groups or a **line** chart for a trend is a *functionality* decision worth marks — but only if you **say why** that chart suits that data.

How to answer exam questions

Read the **command word** first — it sets how much to write. *State/Identify* wants a word or phrase; *Describe* wants features; *Explain* wants reasons and links; *Justify/Evaluate* wants a judgement backed by evidence. Always pull in the **scenario detail** rather than answering in the abstract. The examples below contrast a weak response with a strong one for the same question.

Q1. Identify · 2 marks

Two of the four design tools named in this outcome, and for each state the one thing it plans.

✗ A weak answer
"Mock-ups and IPO charts." — Names two tools but never says what each plans, so it answers only half the question. 0–1 marks.

✓ A strong answer
"An **IPO chart** plans the functionality — what data goes in, the process, and the output. A **mock-up** plans the appearance — the on-screen layout of the dashboard." — Two tools, each with its job. 2 marks.

How the marks are awarded

- 1 mark — naming each correct tool (max 1 for naming).
- 1 mark — stating what each plans (functionality vs appearance, or a correct specific job).
- Accept IPO, mock-up, annotated diagram, query design.

Q2. Distinguish · 4 marks

Between a mock-up and an annotated diagram, using the StepUp dashboard as your example.

✗ A weak answer

"A mock-up is a design and an annotated diagram is a design with notes. They are similar." — Gestures at the difference but gives no clear distinction and no StepUp example. 1 mark.

✓ A strong answer

"A **mock-up** is a layout sketch — for StepUp it shows the title bar, KPI tiles, bar chart and leaderboard table in their positions, with placeholder content. An **annotated diagram** is that same mock-up *with callout notes* explaining each choice — e.g. 'bar chart chosen to compare home groups (correct chart type)', 'title largest for hierarchy'. **Whereas** the mock-up only shows the look, the annotated diagram also justifies it with design principles." — Clear contrast + scenario example for each. 4 marks.

How the marks are awarded

- 1 mark — correct description of a mock-up (layout/appearance).
- 1 mark — correct description of an annotated diagram (mock-up + explanatory notes).
- 1 mark — a clear point of distinction ("whereas…").
- 1 mark — a relevant StepUp example for each.

Q3. Practice question · 3 marks

Construct an IPO chart for the process that turns the daily step file into the ranked home-group leaderboard.

✗ A weak answer

"Input: data. Process: calculate the leaderboard. Output: leaderboard." — Three columns exist, but the process is a single vague verb that hides every step, and the input names no fields. 1 mark.

✓ A strong answer

Input: daily file (StudentID, Date, Steps, Homeroom). **Process:** 1) remove blank/negative Steps; 2) SUM Steps per Homeroom; 3) sort totals highlow; 4) assign rank. **Output:** table of Rank, Homeroom, Total steps. — Named fields, numbered process steps, specific output. 3 marks.

How the marks are awarded

- 1 mark — Input names the actual fields of the file.
- 1 mark — Process lists clear ordered steps (clean sum sort rank).
- 1 mark — Output is the specific ranked leaderboard (not just "data").

Q4. Practice question · 4 marks

The coordinator wants to see only home groups whose total beats 50,000 steps, highest first. Design a database query for this, and name one reason a written query design is useful before building.

✗ A weak answer

"Search for the groups over 50000 and put the biggest first." — The intent is right but no fields, no structure, nothing a developer could build from. 1 mark.

✓ A strong answer

SELECT Homeroom, SUM(Steps) AS Total FROM StepLog GROUP BY Homeroom HAVING SUM(Steps) > 50000 ORDER BY Total DESC; "A written query design is useful because it lets you check the logic and agree the criteria with the client *before* spending time building it." — Fields, grouping, condition, sort + a valid reason. 4 marks.

How the marks are awarded

- 1 mark — correct fields shown (Homeroom + total of Steps).
- 1 mark — the > 50,000 condition (HAVING/criteria).
- 1 mark — sort highestlowest (ORDER BY DESC).
- 1 mark — a valid reason a query design helps before building.